

1 Connectedness and Compactness

1.1 Connected Spaces

Problem 1:

Let $\mathcal{T}$ and $\mathcal{T}'$ be two topologies on $\mathcal{T}$. If $\mathcal{T}' \supset \mathcal{T}$, what does connectedness of X in one topology imply about connectedness in the other?

Claim

If $A \subset X$ is connected in $\mathcal{T}'$, then A will be connected in $\mathcal{T}$.

Proof.

Let $A \subset X$ be connected in $\mathcal{T}'$. This means there are no sets $U, V \in \mathcal{T}'$ s.t. $U \sqcup V = A$. Since $\mathcal{T} \subset \mathcal{T}'$ since if $U \in \mathcal{T} \implies U \in \mathcal{T}'$ this means that there are also no $U, V \in \mathcal{T}$ s.t. $U \sqcup V = A$. ■

Problem 2:

Let $\{A_n\}$ be a sequence of connected subspaces of X , such that $A_n \cap A_{n+1} \neq \emptyset$ for all n . Show that $\bigcup A_n$ is connected.

Proof.

Assume for contradiction that $A = \bigcup A_n$ is disconnected. Thus, there exists nonempty open disjoint sets U, V s.t. $A = U \cup V$. Since U and V are not empty this implies that there exists an $n \in \mathbb{N}$ s.t. $A_n \subseteq U$ and $A_{n+1} \subseteq V$. Since $U \cap V = \emptyset \implies A_n \cap A_{n+1} = \emptyset \implies \Leftarrow$ Thus, A is connected. ■

Problem 3:

Let A_α be a collection of connected subspaces of X ; let A be a connected subspace of X . Show that if $A \cap A_\alpha \neq \emptyset$ for all α , then $A \cup (\bigcup A_\alpha)$ is connected.

Proof.

Assume for contradiction that $O = A \cup (\bigcup A_\alpha)$ is not connected.

Thus, there exists nonempty open disjoint sets U, V s.t. $O = U \cup V$.

Since U and V are not empty this implies there exists an α s.t. $A_\alpha \subseteq U$ and $A \subseteq V$.

This is a contradiction since $A \cap A_\alpha \neq \emptyset$.

Thus, O is connected. ■

Problem 4:

Show that if X is an infinite set, it is connected in the finite complement topology

Proof.

Assume X is an infinite set.

Assume for contradiction that X is disconnected.

Thus, there exists nonempty open disjoint sets U, V s.t. $X = U \sqcup V$.

Thus, $X \setminus U$ and $X \setminus V$ are finite.

Consider:

$$(X \setminus U) \cup (X \setminus V) = X \setminus (U \cap V) = X \setminus \emptyset = X$$

This is a contradiction since X is infinite which we have shown is a union of two finite sets. Therefore, X is connected. ■

Problem 6:

Let $A \subseteq X$. Show that if C is a connected subspace of X that intersects both A and $X \setminus A$, then C intersects ∂A .

Proof.

Assume C is a connected subspace of X that intersects both A and $X \setminus A$.

Consider $\partial A = \overline{A} \cap \overline{X \setminus A}$.

Assume for contradiction that $\partial A \cap C = \emptyset$.

Since $X = \overline{A} \cup \overline{X \setminus A}$ we know that for $x \in C$ that $x \in \overline{A}$ or $x \in \overline{X \setminus A}$ not both since $C \cap \partial A = \emptyset$.

Consider $C = (C \cap \overline{A}) \cup (C \cap \overline{X \setminus A})$.

Consider $(\overline{A} \cap \overline{X \setminus A}) \cap (X \setminus \overline{A} \cup X \setminus \overline{X \setminus A}) = X \setminus (\overline{A} \cup \overline{X \setminus A}) = X \setminus X = \emptyset$.

Thus, this is a separation of C which is a contradiction. ■

Problem 9:

Let A be a proper subset of X , and let B be a proper subset of Y . If X and Y are connected, show that

$$(X \times Y) \setminus (A \times B)$$

is connected.

Proof.

Let $A \subsetneq X$ and $B \subsetneq Y$.

Let $a \notin A$ and $b \notin B$ consider $(\{a\} \times Y) \cup (X \times \{b\}) = C$. This is connected since these spaces share (a, b) and one point product with space X, Y is homeomorphic to X, Y . Thus, it is connected.

Since $(\{x\} \times Y)$ and $(X \times \{y\})$ for $x \notin A$ and $y \notin B$ intersect C . By problem 3 we get that

$$\left(\bigcup_{x \notin A} \{x\} \times Y \right) \cup \left(\bigcup_{y \notin B} X \times \{y\} \right) = (X \times Y) \setminus (A \times B)$$

is connected. ■

Problem 10:

Let $\{X_\alpha\}_{\alpha \in J}$ be an indexed family of connected spaces; let X be the product space

$$X = \prod_{\alpha \in J} X_\alpha$$

Let $\mathbf{a} = (a_\alpha)$ be a fixed point of X .

(a)

Given any finite subset K of J , let X_K denote the subspace of X consisting of all points $\mathbf{x} = (x_\alpha)$ such that $x_\alpha = a_\alpha$ for $\alpha \notin K$. Show that X_K is connected.

Proof.

Assume for contradiction that X_K is disconnected. Thus, consider the projection map from X to $B = X_{\alpha_1} \times \cdots \times X_{\alpha_n}$ s.t. $\alpha_i \in K$. Since X_{α_i} is connected we know that the product of all of them is connected thus consider $\pi(X_K) = \pi(U) \cup \pi(V) = B$. Since $U \cap V = \emptyset$. If $\pi(u) = \pi(v)$ for $u \in U$ and $v \in V$. This implies that $u = v$ since there are already equal for all things not in K but then when you project it it shows they are equal for also K . This is a contradiction of $U \cap V = \emptyset$. Thus, $\pi(U) \cap \pi(V) = \emptyset$. Since $X_K \cong \pi(X_K)$ since we just have the times one point upfront this means that $\pi(U), \pi(V)$ are open thus this is a contradiction of B being connected, since $\pi(X_K) = B$. ■

(b)

Show that the union Y of the spaces X_K is connected.

Proof.

Consider $\{\mathbf{a}\}$. This is trivially connected, and $\{\mathbf{a}\} \cap X_K = \{\mathbf{a}\}$ for all $K \subset J$ s.t. K is finite. Thus, by problem 3 we get that Y is connected. ■

(c)

Show that X equals the closure of Y ; conclude that X is connected.

Proof.

Since X is closed and contains Y this implies that $\overline{Y} \subset X$.

Let $x \in X$. Let U be a neighborhood of x . Thus, there is a basis element s.t. for

$$K = \{\alpha_1, \dots, \alpha_n\} \subseteq J$$

$$B = U_{\alpha_1} \times \dots \times U_{\alpha_n} \times \left(\prod_{\alpha \notin K} X_\alpha \right) \subseteq U$$

Consider $y = (y_\alpha) \in B$ s.t. $y_\alpha = a_\alpha$ for $\alpha \notin K$ and $y_{\alpha_i} \in U_{\alpha_i}$ for $\alpha_i \in K$.

Thus, $y \in Y$. Thus, $X = \bar{Y}$. Thus, $Y \subseteq X \subseteq \bar{Y} \implies X$ is connected. ■

Problem 11:

Let $p : X \rightarrow Y$ be a quotient map. Show if each set $p^{-1}(\{y\})$ is connected, and if Y is connected then X is connected.

Proof.

Assume $p^{-1}(\{y\})$ is connected for all $y \in Y$ and Y is connected.

Assume for contradiction that $X = U \sqcup V$ for U, V open.

Thus, for $y \in Y$, $p^{-1}(\{y\})$ is either in U or V .

Thus, since p is surjective and the preimages of single points only live in either U or V but not both we can partition Y into sets S, T s.t. $p^{-1}(S) = U$ and $p^{-1}(T) = V$. This is a contradiction since Y is connected. ■

Problem 12:

Let $Y \subseteq X$; let X and Y be connected. Show that if A and B form a separation of $X \setminus Y$, then $Y \cup A$ and $Y \cup B$ are connected

Proof.

Assume $Y \cup A$ is disconnected then $Y \cup A = U \cup V$ for nonempty disjoint open sets U, V .

Consider:

$$\begin{aligned} X &= Y \cup A \cup B \\ &= U \cup V \cup B \end{aligned}$$

Now consider $U \cap (V \cup B)$

$$\begin{aligned} U \cap (V \cup B) &= (U \cap V) \cup (U \cap B) \\ &= \emptyset \cup (U \cap B) \end{aligned}$$

Since $U \subset Y \cup A$ it can only have things from Y or A thus since $B \cap A = \emptyset$ and $B \cap Y = \emptyset$. This means that $U \cap B = \emptyset$

$$= \emptyset$$

This is a contradiction of X being connected. ■